Then, 10 (54 − x) + 22x = 780 ⇔ 12x = 240 ⇔ x = 20.

∴ Bigger part = (54 − x) = 34.

122. Let the three parts be A, B and C.

Let $\frac{A}{2} = \frac{B}{3} = \frac{C}{4} = x$.

Then, A = 2x, B = 3x and C = 4x.

So, A : B : C = 2 : 3 : 4.

∴ Largest part = $\left(243 \times \frac{4}{9}\right)$ = 108.

123. Let the four numbers be A, B, C and D.

Let $A + 3 = B - 3 = 3C = \frac{D}{3} = x$.

Then, $A = x - 3$, $B = x + 3$, $C = \frac{x}{3}$ and $D = 3x$.

$A + B + C + D = 64 \Rightarrow (x-3)+(x+3)+\frac{x}{3}+3x = 64$

$\Rightarrow 5x + \frac{x}{3} = 64 \Rightarrow 16x = 192 \Rightarrow x = 12.$

Thus, the numbers are 9, 15, 4 and 36.

∴ Required difference = (36 − 4) = 32.

124. Let the numbers be a, b and c. Then, $a^2 + b^2 + c^2 = 138$ and $(ab + bc + ca) = 131$.

$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) = 138 + 2 \times 131 = 400$

$\Rightarrow (a + b + c) = \sqrt{400} = 20.$

125. A : B = 2 : 3 and B : C = 5 : 3 = $\frac{3}{5} \times 5 : \frac{3}{5} \times 3 = 3 : \frac{9}{5}$.

So, A : B : C = 2 : 3 : $\frac{9}{5}$ = 10 : 15 : 9.

∴ Second number = $\left(136 \times \frac{15}{34}\right) = 60.$

126. Let the numbers be x, y and z.

Then, $x + y = 73$, $y + z = 77$ and $3x + z = 104$.

∴ $y = 73 - x$, $z = 77 - y = 77 - (73 - x) = 4 + x$.

∴ $3x + 4 + x = 104$ ⇔ $4x = 100$ ⇔ $x = 25$.

$y = (73 - 25) = 48$ and $z = (4 + 25) = 29$.

∴ Third number = 29.

127. Let two numbers be a and b

Given $ab = 5$ and $a = \frac{3}{2}$

$\Rightarrow b = \frac{5}{a}$

$b = \frac{5}{\frac{3}{2}} = \frac{5 \times 2}{3} = \frac{10}{3}$

∴ Required sum of $a + b = \frac{3}{2} + \frac{10}{3}$

LCM of 2 and 3 = 6

$= \frac{9+20}{6} = \frac{29}{6} = 4\frac{5}{6}$

128. Let the positive integers be a and b where $a > b$.

According to the question,

$a^2 + b^2 = 100$ ....(i)

$a^2 - b^2 = 28$ ....(ii)

By adding (i) and (ii), we get

$\therefore a^2 + b^2 + a^2 - b^2 = 100 + 28$

$\Rightarrow 2a^2 = 128$

$\Rightarrow a^2 = \frac{128}{2} = 64$

$\therefore a = \sqrt{64} = 8$

From equation (i).

$8^2 + b^2 = 100$

$\Rightarrow b^2 = 100 - 64 = 36$

$\Rightarrow b = \sqrt{36} = 6$

$\therefore a + b = 8 + 6 = 14$

129. Let the numbers be a and b where $a > b$.

According to the question,

$a + b = 37$ and $a^2 - b^2 = 185$

$\Rightarrow (a+b)(a-b) = 185$

$\Rightarrow 37(a-b) = 185$

$\Rightarrow a - b = \frac{185}{37} = 5$

130. Let the total number of eggs bought be a.

10% of eggs are rotten.

∴ Remaining eggs

$= a - 10\% \text{ of } a = a - \frac{10a}{100} = \frac{100a - 10a}{100} = \frac{90a}{100} = \frac{9a}{10}$

Man gives 80% of $\frac{9a}{100}$ eggs to his neighbour

$= \frac{80}{100} \times \frac{9a}{10} = \frac{72a}{100}$

Remaining eggs = $\frac{9a}{10} - \frac{72a}{100} = \frac{90a - 72a}{100} = \frac{18a}{100} = \frac{9a}{50}$

According to the question.

$\frac{9a}{50} = 36 \Rightarrow 9a = 36 \times 50$

$\Rightarrow a = \frac{36 \times 50}{9} = 200$

Hence the total number of eggs. bought be 200.

131. Let the numbers be a and b.

According to the question.

$a + b = 75$

$a - b = 25$

$\because (a+b)^2 - (a-b)^2 = 4ab$

$\Rightarrow 75^2 - 25^2 = 4ab$

$4ab = (75+25)(75-25)$

$\left[\because a^2 - b^2 = (a+b)(a-b)\right]$

$\Rightarrow 4ab = 100 \times 50$

$\Rightarrow ab = \frac{100 \times 50}{4} = 1250$

132. Let the five successive odd number be,

$x, x+2, x+4, x+6, x+8$